

Design Thinking

CS571: Building User Interfaces

Cole Nelson & Yuhang Zhao

What will we learn today?

- What is design thinking?
- How do we empathize with users?
- How can we turn this empathy into insight?
- How to generate design ideas?

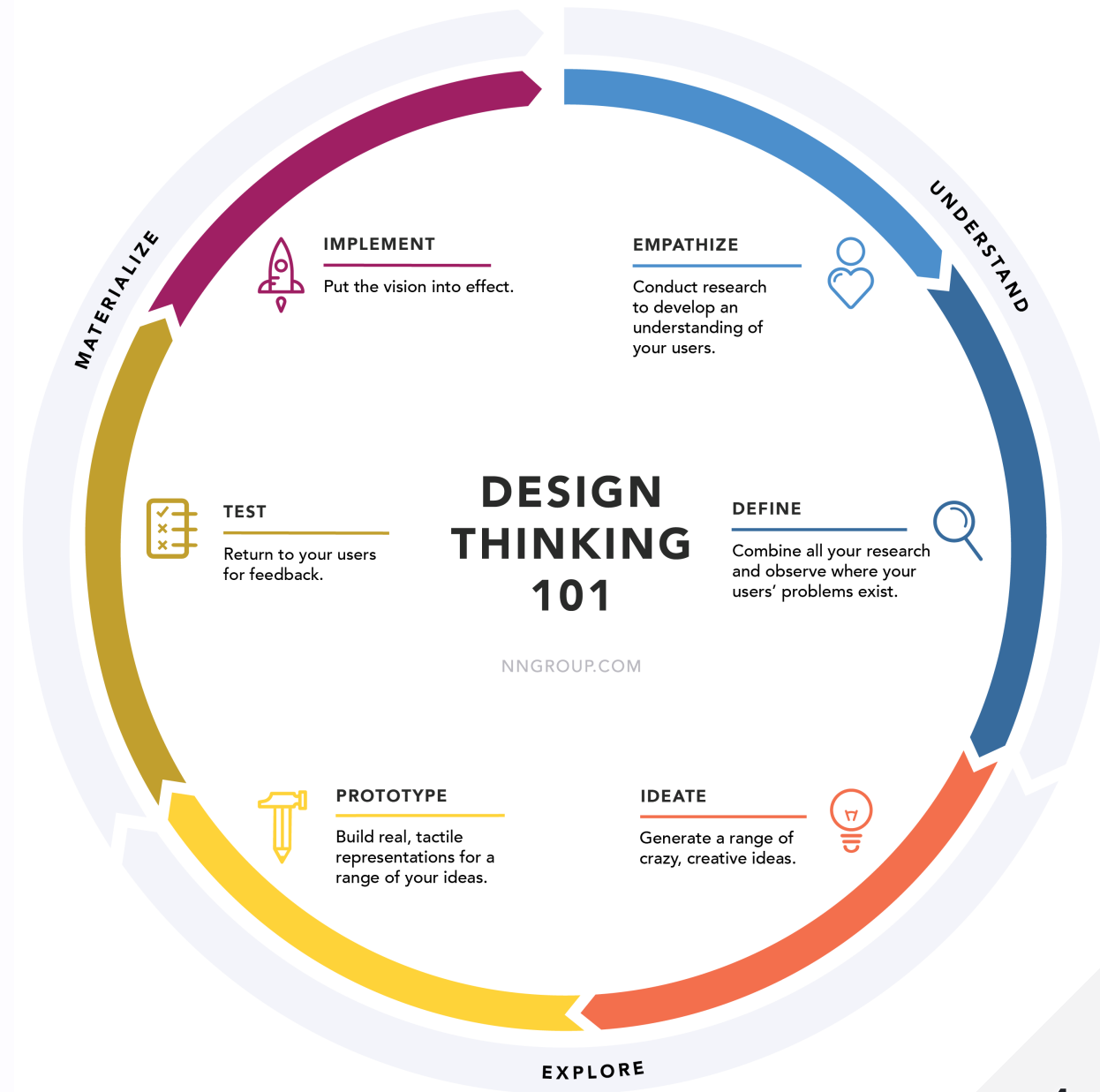


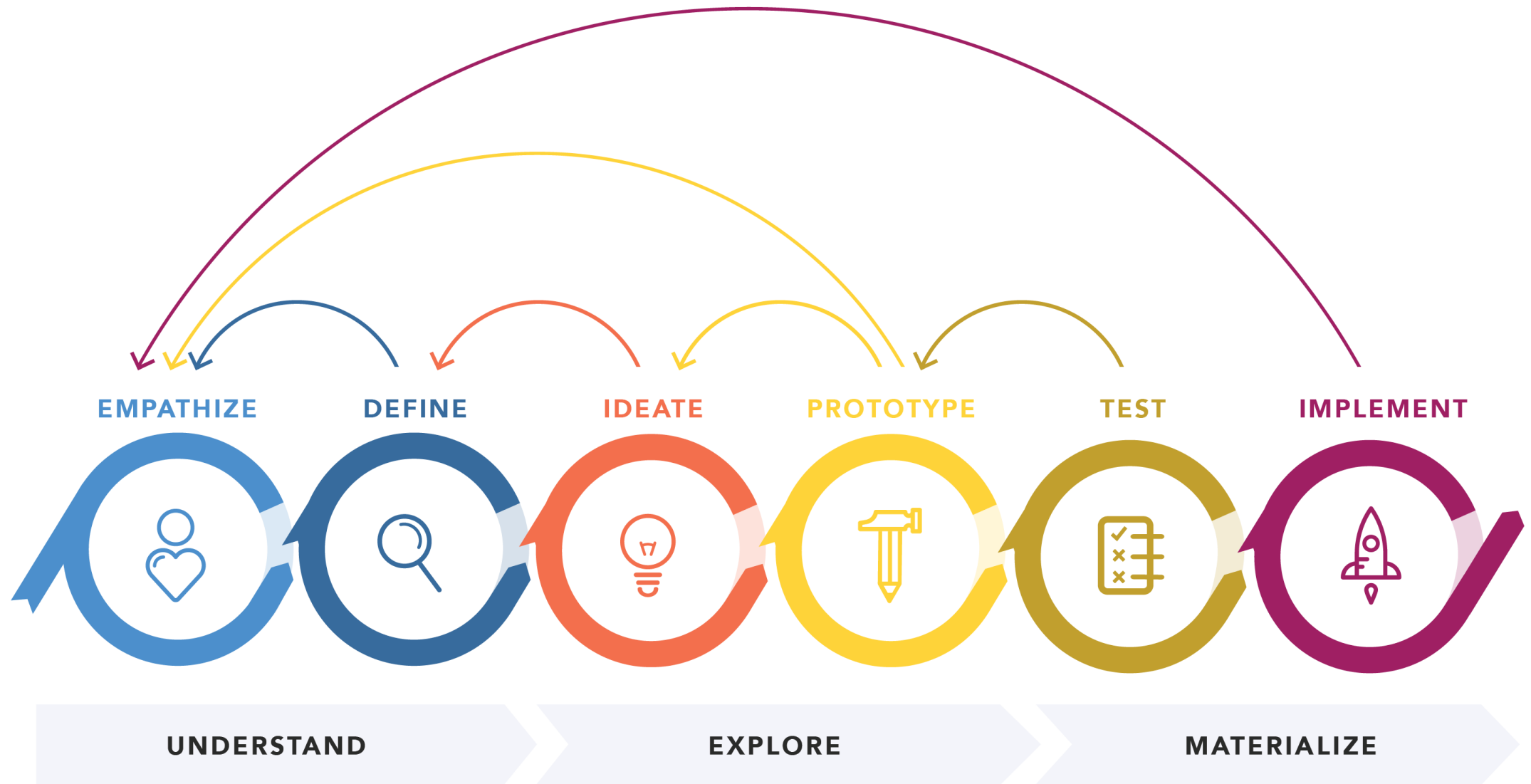
NN/g Design Thinking 101

Design Thinking

1. An **approach**: a hands-on, user-centric approach to innovative problem solving
2. A **process**: six phases

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Empathize

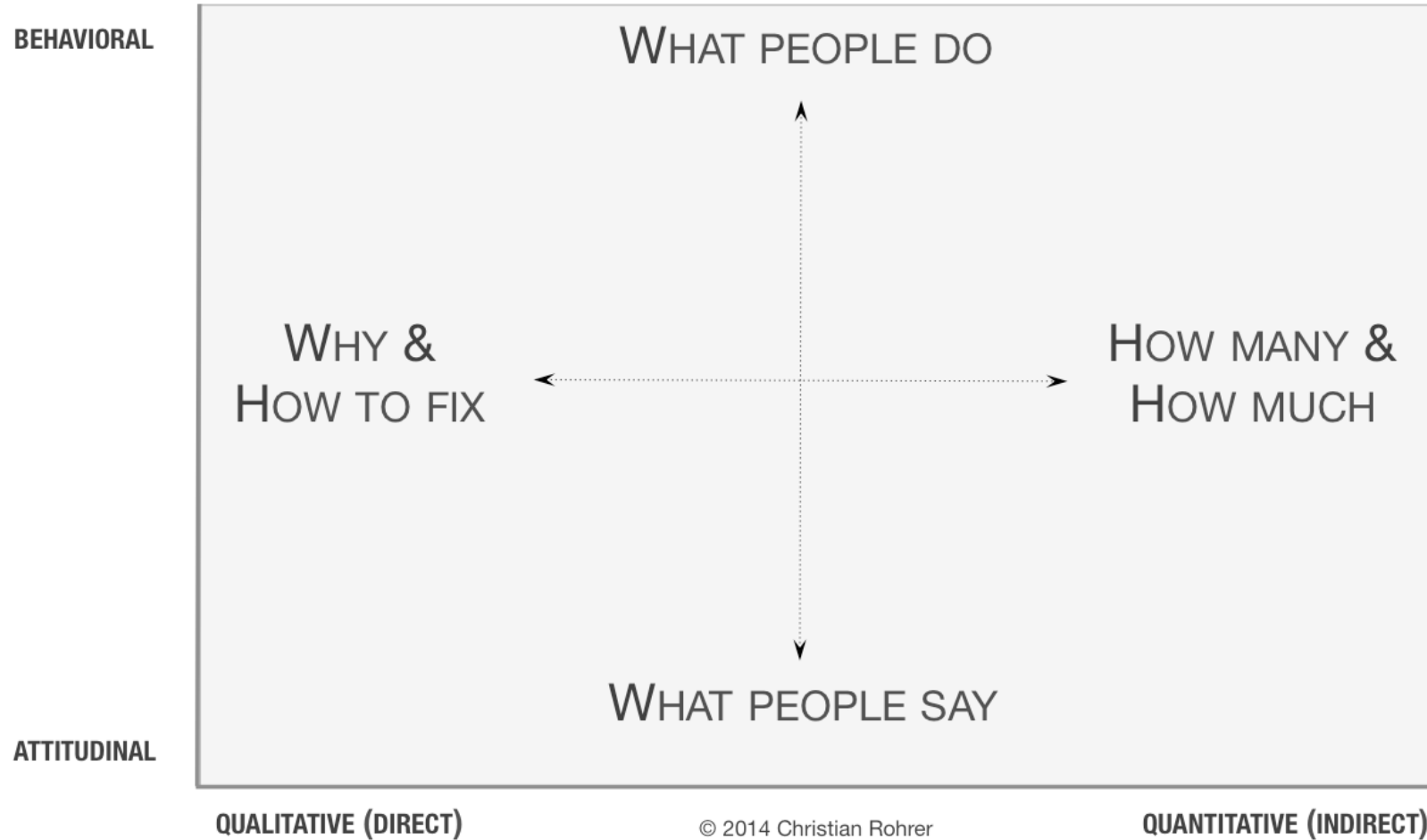
Conduct research to develop an understanding of your users.

How do we empathize with users?

Understanding user needs, preferences, and expectations by studying what users do, say, think, and feel.

There are many methods to build empathy.

QUESTIONS ANSWERED BY RESEARCH METHODS ACROSS THE LANDSCAPE



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Think-Aloud

The simplest and most powerful method for empathy



Nomensa

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Conducting a Think-Aloud

1. Identify representative users.
2. Ask users to perform representative tasks.
3. Observe and record what they say and do, noting obstacles, pain points, and failures.

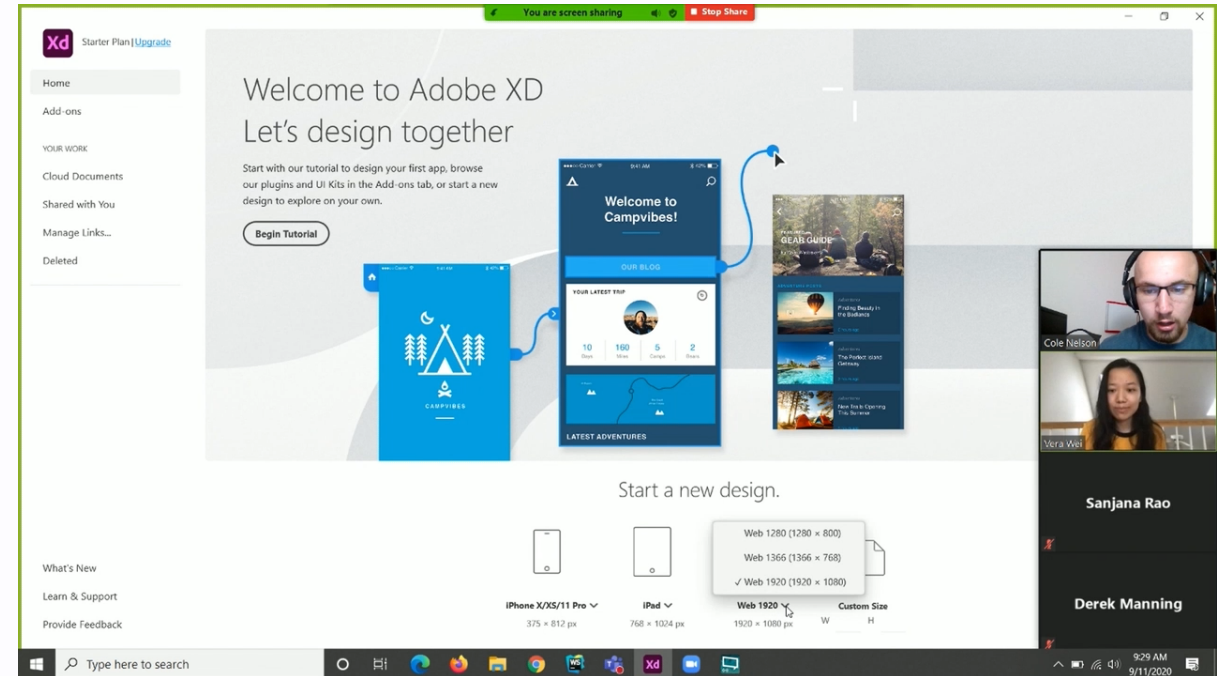
A think-aloud can be *moderated* or *unmoderated*.

A think-aloud can be done in *field* or in *lab*.

A think-aloud can be done *presently* or *retrospectively*.

In-Class Activity Part I

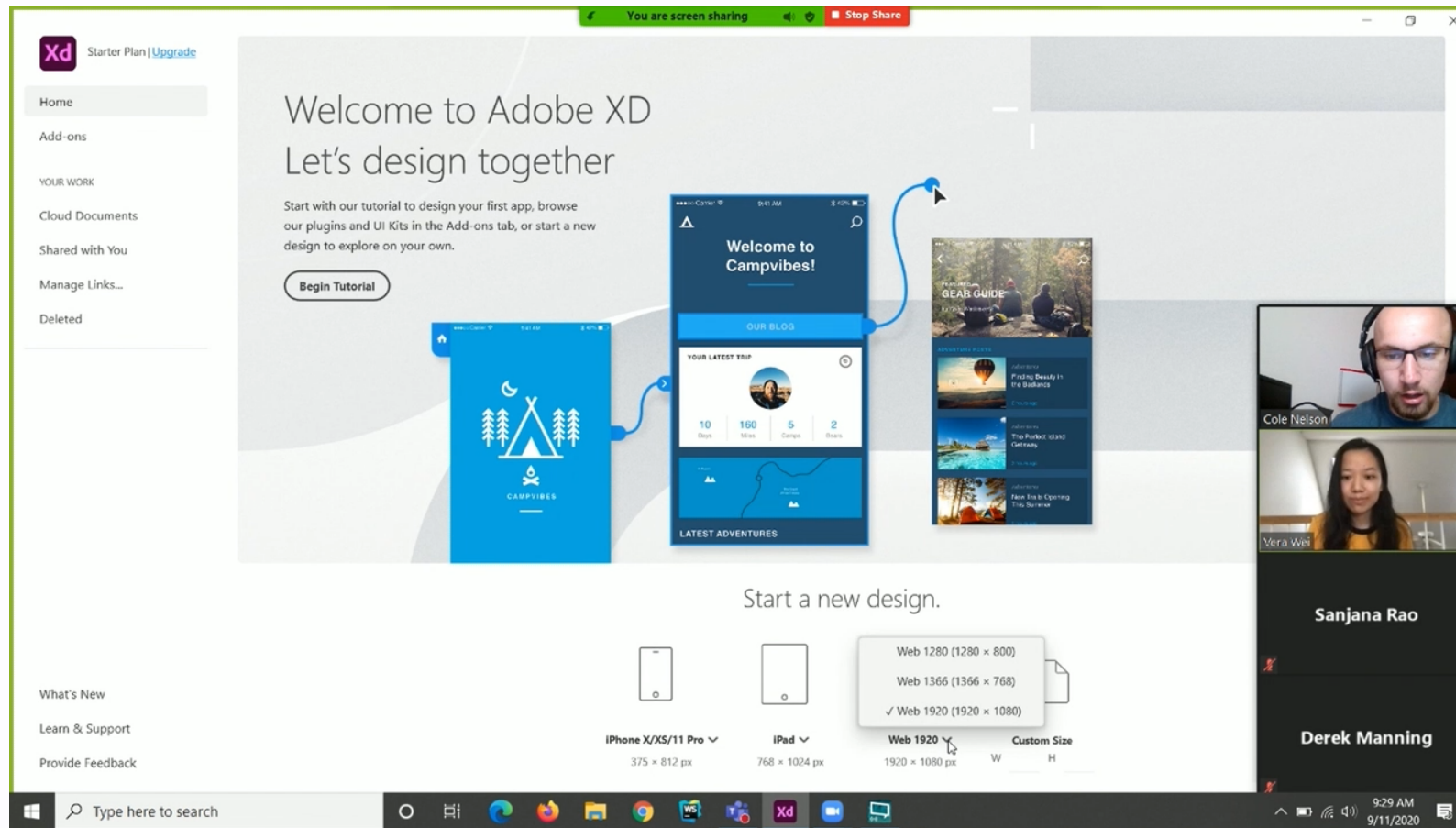
*A think-aloud session
with a novice user
performing a set of tasks
in Adobe XD.*



What to do

Grab a post-it note on [this Google Drawings canvas](#). Write down one key observation on the post-it note. 🖐️ *Repeat for additional observations.*





Think-Aloud Demo

Define

Combine all your data and derive where your users' problems exist.

What does that mean?

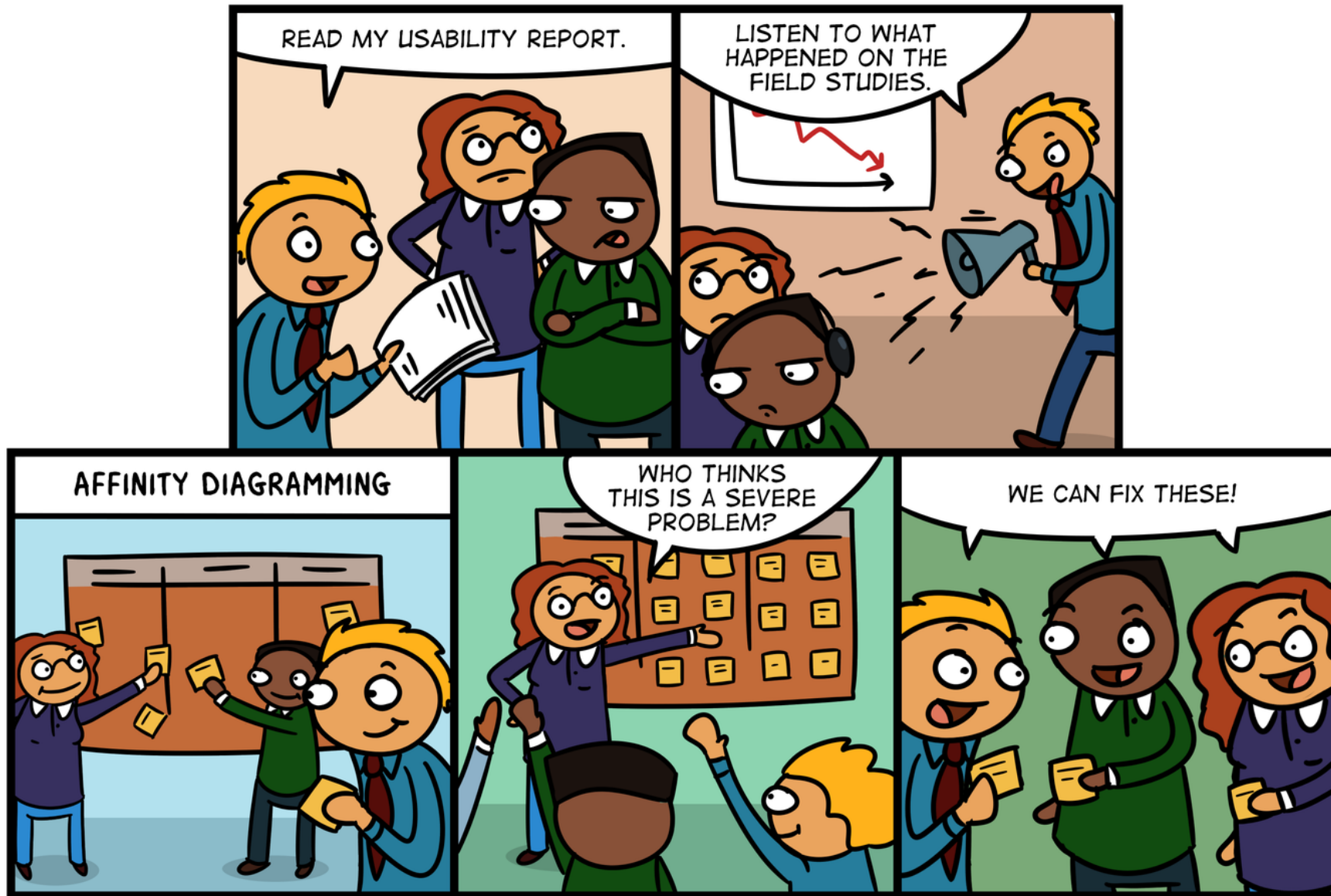
- Gathering all findings — **gather**
- Consolidating, categorizing, distilling — **analyze**
- Translate into insight — **recommend**

But how do we go about this?

- Again, there are many methods for analysis.
- The simplest and most powerful method is **affinity diagramming**.

Affinity Diagramming

- **Definition:** Organizing data into clusters based on "affinity."
- It helps you make sense of qualitative, messy data.
- Also known as *affinity mapping*, *collaborative sorting*, *snowballing*.
- Used across the board in creative, generative industries.



NN/g Affinity Diagramming

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NNGROUP.COM **NN/g**

How do we go about it?

- **Step 1:** Start with an initial set of categories
- **Step 2:** Sort notes into these categories
- **Step 3:** Add subcategories/consolidate as needed.
- **Step 4:** Present each category
- **Step 5:** Rank categories in *severity*, combining importance, prevalence, frequency

Pro Tip: Steps 2–3 will likely be iterative.

Pro Tip: Assign team members to categories who will be responsible for all the sorting and presentation of the categories

UX Collective

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In-Class Activity Part II

Affinity Diagramming in [Google Drawings Canvas](#)

Your findings will serve as *design insight*.

Ideate

Generate a range of crazy, creative ideas.

Ideation

Definition: An active, creative, exploratory, highly iterative, fast-moving collaborative process for forming ideas for design.

Ideation can be done *individually* or *collaboratively*.

Ideation has two stages:

Idea creation → Critiquing

Key Considerations

- Workspace
- Team
- Process
- Rules of engagement
- Method of capturing ideas

GlassDoor



IDEO's Rules of Engagement

1. Defer judgement.
2. Encourage wild ideas.
3. Build on the ideas of others.
4. Stay focused on the topic.
5. One conversation at a time.
6. Be visual.
7. Go for quantity.

What do we do with ideas?

Sketching. Storyboarding.

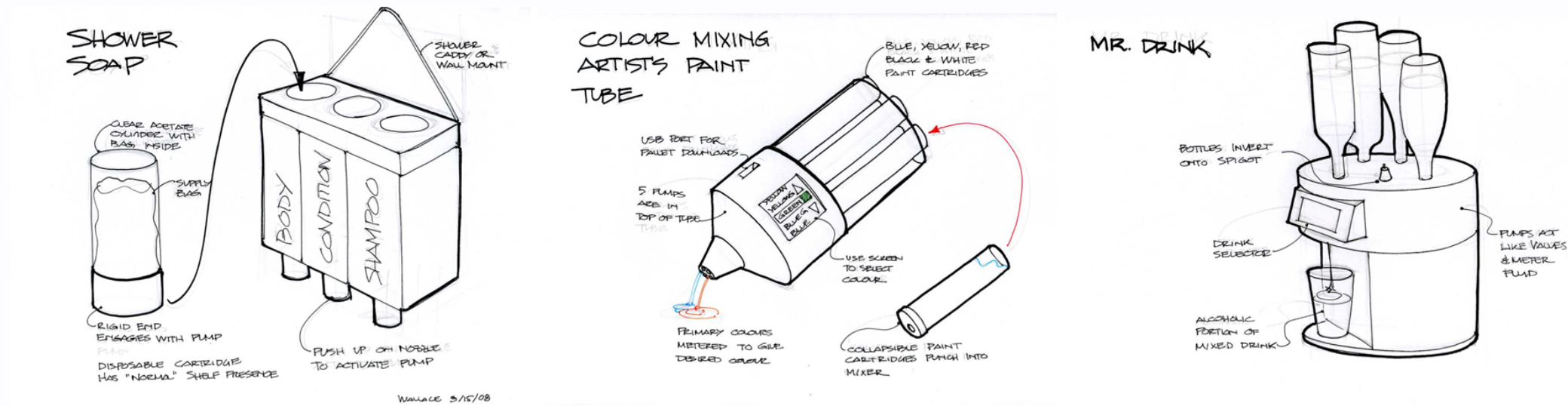
Sketching & storyboarding exhibit *conceptual designs*.

What is Conceptual Design?

Definition: An abstract characterization of the context, use, or experience with an envisioned design solution that highlights the main premise of the solution.

Sketching

Definition: A sketch is a quick and rough drawing that gives a general outline of an idea.

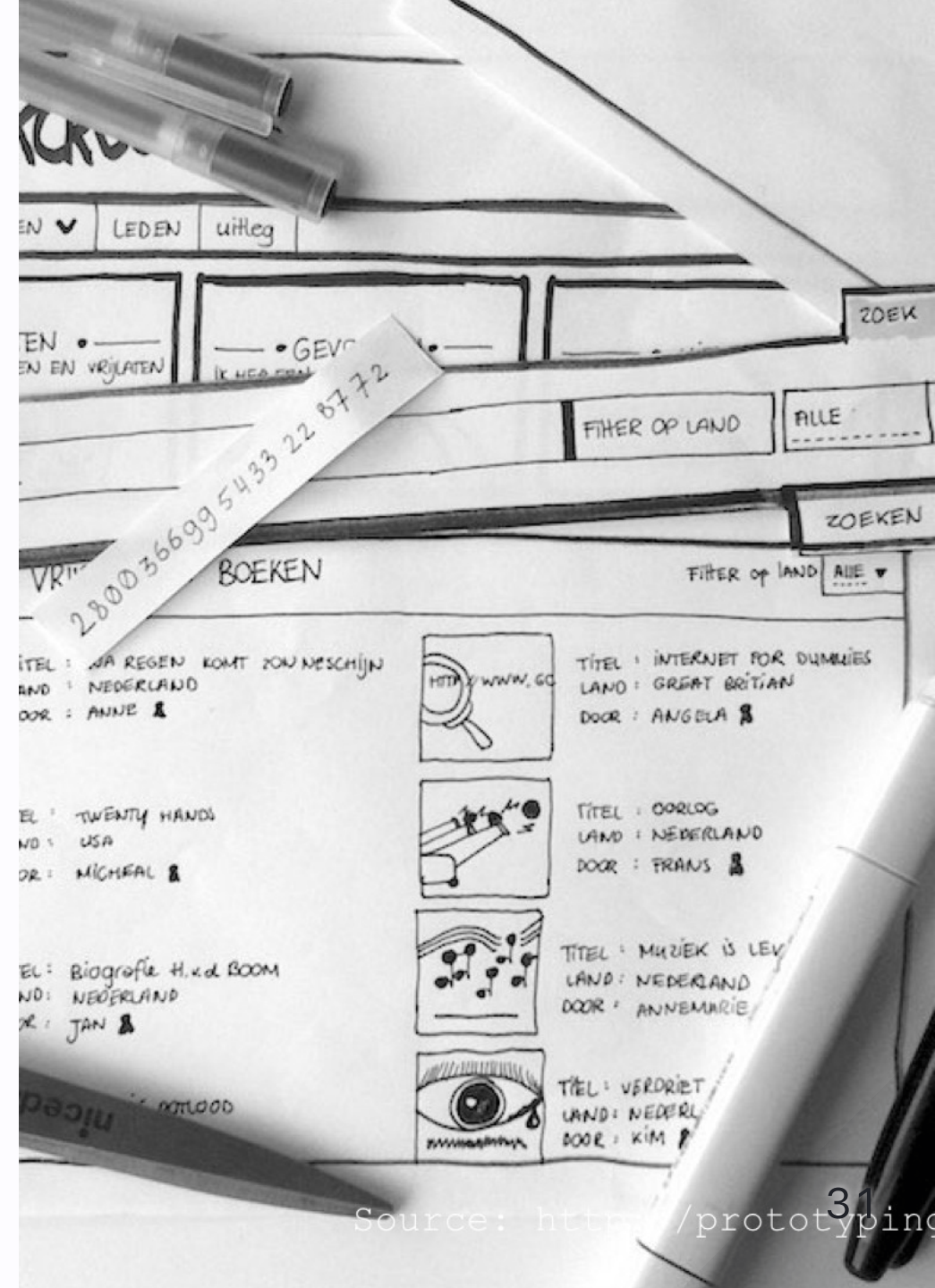


Sketching

- Everyone can sketch.
- Sketching is more effective than words for most ideas.
- Quick and inexpensive sketches do not inhibit exploration.
- Sketches are disposable.

Buxton, 2007

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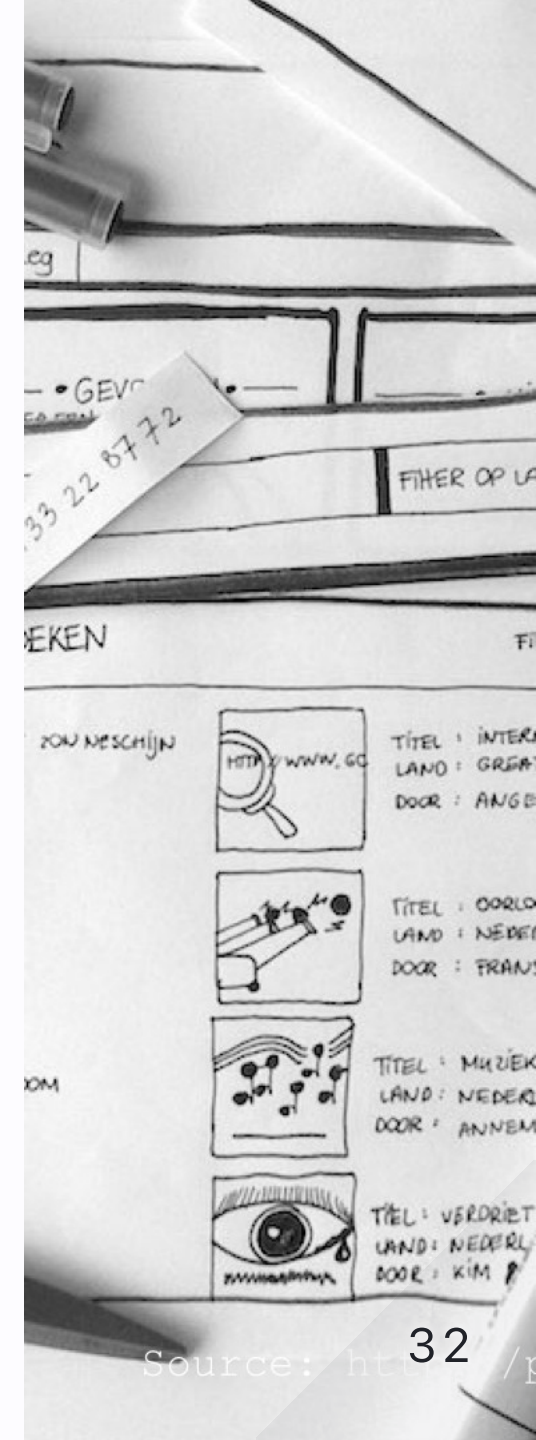


Source: <http://prototyping31>

Sketching

- Sketches are made just-in-time, in-the-moment, when needed.
- Sketches should be plentiful, entertain a large number of ideas, and include multiple sketches of each idea.
- Textual annotations can explain what is going on in the sketch.

Buxton, 2007





YOU CAN DRAW,
IT IS NOT ABOUT
BEING ARTISTIC!



JUST START IT, YOU'LL
BECOME MORE CONFIDENT
OVER TIME!

SKETCHING

UX Knowledge Base Sketch #52
FOR UX DESIGNERS

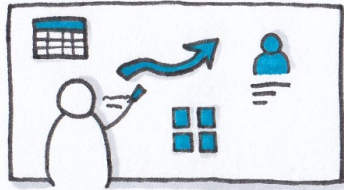


IN CASE OF USER INTERFACES:
VARY THE FIDELITY/
DETAIL LEVEL BASED ON

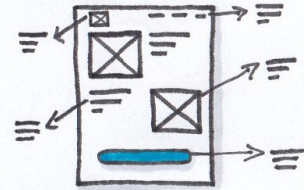
YOUR
GOAL
(DELIVERABLE?
QUICK CONCEPT?)

YOUR
AUDIENCE
(CLIENT? TEAM?
YOURSELF?)

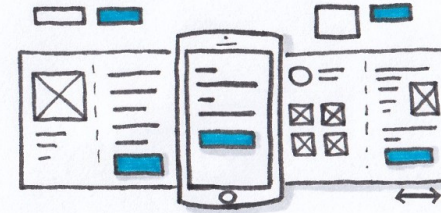
TOOLS:
JUST GRAB THE NEAREST
PEN & PAPER!



①. FACILITATING MEETINGS &
DESIGN WORKSHOPS,
PROJECT PLANNING



②. WIREFRAMING
DON'T FORGET:
ANNOTATIONS ARE GREAT!



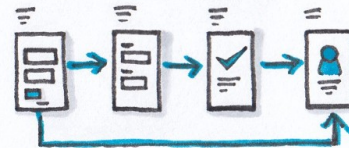
③. PAPER PROTOTYPING -
VALIDATING IDEAS,
TESTING OUT CONCEPTS



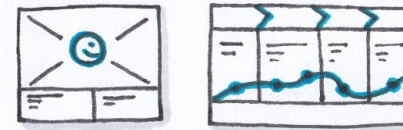
④. IDEATION
QUICK IDEA GENERATION
(E.G. DURING A DESIGN SPRINT,
OR JUST ON YOUR OWN)



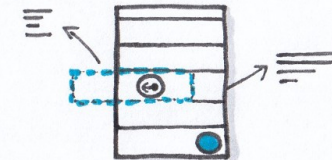
⑤. TEAMWORK, ANY KIND OF
COLLABORATION
(E.G.: "TALKING SKETCHES")



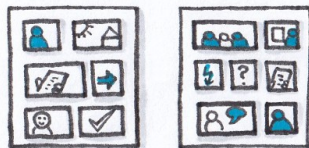
⑥. USER FLOWS
SITEMAPS
INFORMATION ARCHITECTURE



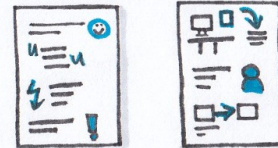
⑦. MAPPING: EMPATHY MAP,
JOURNEY MAP,
PRODUCT ROADMAP ETC.



⑧. UI ANIMATIONS
WHAT CHANGES, HOW,
WHAT THE TRIGGER IS



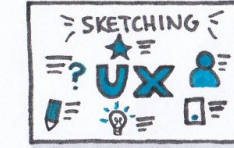
⑨. STORYBOARDING
VALIDATING
ASSUMPTIONS



⑩. APPLYING ICONS, VISUALS
IN UX RESEARCH NOTES
(E.G. USER INTERVIEW/
CONTEXTUAL INQUIRY)



⑪. DOCUMENTATION,
PRESENTATION
TO CLARIFY &
TO MAKE IT MORE ENGAGING



⑫. SKETCHNOTING
- CONFERENCE TALKS
- BOOKS
- MEETINGS

Medium

Storyboarding

Definition: A sequence of visual frames that illustrate user interaction with the envisioned system, capturing social, environmental, and technical factors that shape user experience.

Take your sketches (in context!) and impose them on a storyboard.

STORYBOARD



- MAKES NOTE OF SUPPLIES NEEDED ON CLIPBOARD
- PHYSICAL INVENTORY

PERSONA:

CORPORATE BUYER,
JAMES



- SELECTS ITEMS FROM FAVORITES LIST
- USES DESKTOP & SUPPLY LIST AS TOOL

SCENARIO:

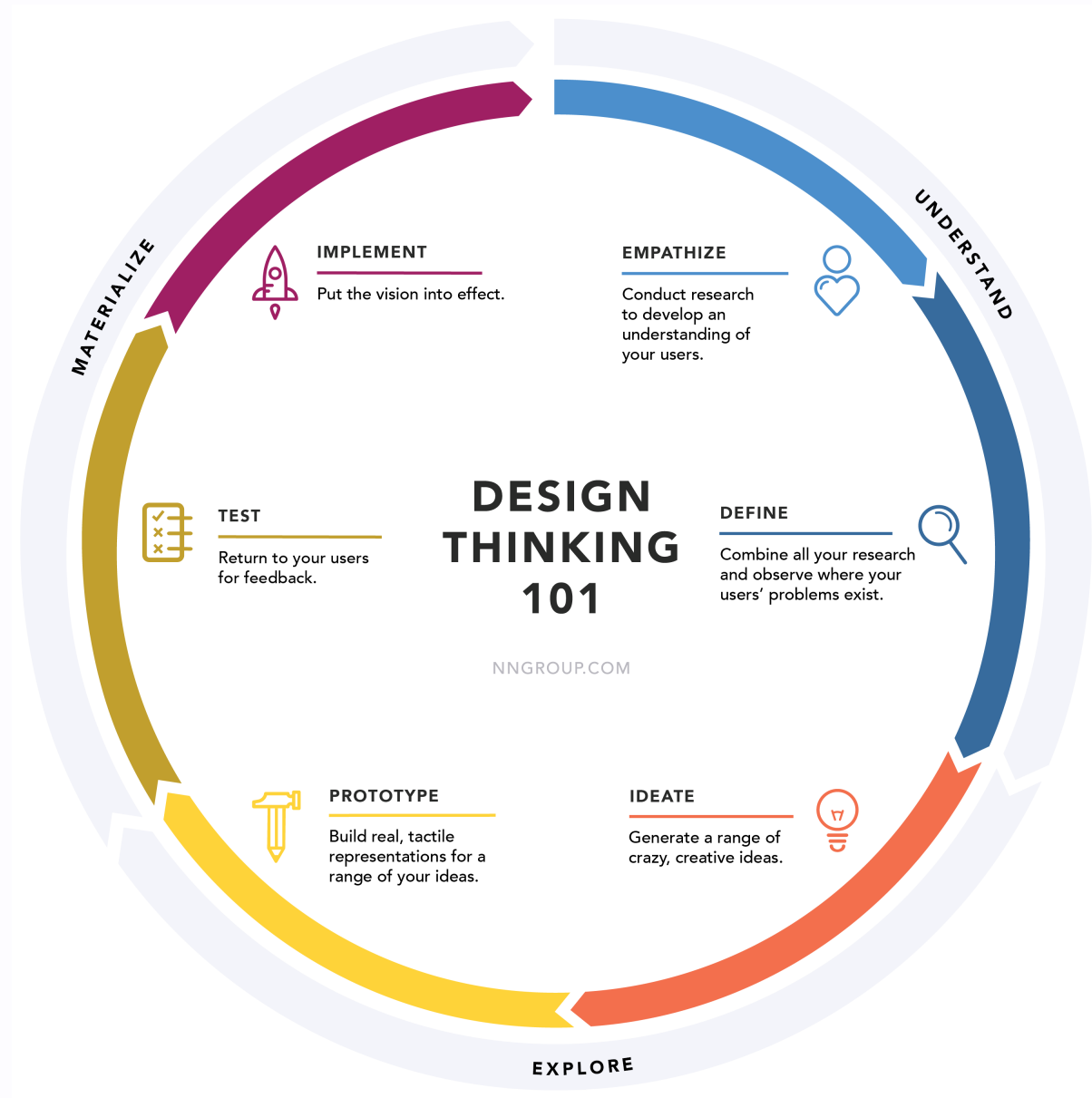
REPLENISH OFFICE SUPPLIES



- RECEIVES SHIPMENT WINDOW W/ORDER SUBMISSION
- SETS PLAN FOR RESTOCK

NN/g: Storyboards Help Visualize UX Ideas

- Prototype
- Test
- Implement



What did we learn today?

- What design thinking is.
- How we can empathize with users.
- How we can turn this empathy into insight.
- How we can generate and present our ideas.

Next Week

JavaScript Bootcamp!

Questions?